

HAOSHENG ZHOU

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EMPLOYMENT

Princeton University, Department of Operations Research and Financial Engineering (ORFE)

Instructor

Aug. 2026 - Present

EDUCATION

University of California, Santa Barbara (UCSB), Department of Statistics and Applied Probability

Ph.D. in Statistics and Applied Probability, emphasis in Financial Mathematics;

Sept. 2022 - June 2026

Dissertation: *Stochastic Differential Games: Theory, Adversarial Interactions, and Learning Methods*

Advisor: Prof. Ruimeng Hu

New York University (NYU), CIMS (Courant Institute)

M.S. in Mathematics;

Sept. 2020 - May 2022

Peking University (PKU), EECS Department

B.S. in Data Science and Big Data Technology, concentration in Internet Finance;

Sept. 2016 - July 2020

RESEARCH INTERESTS

- Stochastic control, stochastic differential games, mean-field control and games.
- Stochastic modeling of adversarial interactions, including deception and counter-deception.
- Deep and reinforcement learning architectures and algorithms for solving multi-agent games.
- Mathematical modeling and analysis in finance, biology, and social science.

PUBLICATIONS & PREPRINTS

- [10] H. Zhou, R. Hu: *Optimal Design of Stealthy Attacks in Partially Observed Linear Systems: A Likelihood-Based Approach*. arXiv:2605.05545, 2026.
- [9] Y. Kim, H. Zhou, A. Benvenuti, R. Hu, M. Hale: *Deception in Linear-Quadratic Control*. arXiv:2604.00227, 2026.
- [8] R. Hu, D. Ralston, X. Yang, H. Zhou: *Strategic Inference in Stackelberg Games: Optimal Control for Revealing Adversary Intent*. arXiv:2510.05641, 2025.
- [7] M. Zhou, H. Zhou, R. Hu: *Learning Mean-Field Games through Mean-Field Actor-Critic Flow*. arXiv:2510.12180, 2025.
- [6] R. Hu, J. Long, H. Zhou: *Finite-Agent Stochastic Differential Games on Large Graphs: II. Graph-Based Architectures*. Accepted for publication in **Communications in Nonlinear Science and Numerical Simulation**, 2026.
- [5] H. Zhou, D. Ralston, X. Yang, R. Hu: *Adversarial Decision-Making in Partially Observable Multi-Agent Systems: A Sequential Hypothesis Testing Approach*. **IEEE Transactions on Control of Network Systems**, 2026.
- [4] H. Zhou, D. Ralston, X. Yang, R. Hu: *Integrating Sequential Hypothesis Testing into Adversarial Games: A Sun Zi-Inspired Framework*. In **Proceedings of IEEE Conference on Decision and Control**, 2025.
- [3] R. Hu, J. Long, H. Zhou: *Finite-Agent Stochastic Differential Games on Large Graphs: I. The Linear-Quadratic Case*. **Applied Mathematics and Optimization (AMO)**, 92(2), 1–50, 2025.
- [2] H. Zhou, W. Lin, S.R. Labra, S.A. Lipton, J.A. Elman, N.J. Schork, A.V. Rangan: *Detecting Boolean Asymmetric Relationships with a Loop Counting Technique and its Implications for Analyzing Heterogeneity within Gene Expression Datasets*. **IEEE ACM Transactions on Computational Biology and Bioinformatics**, 22(1), 27–38, 2024.

- [1] H. Zhou, J. Tu, Y. Kong: *Approximation Algorithms and PTAS of the Minimum Dominating Set and the 3-Path Vertex Cover Problem on Unit Disk Graph*. Bachelor Thesis (non-peer-reviewed), 2020.

HONORS & AWARDS

Nov. 2025: JMM Graduate Student Travel Grant.
Oct. 2025: UCSB Doctoral Student Travel Grant.
July 2025: SIAM Student Travel Awards (SIAM FM 25).
Apr. 2025: UCSB Graduate Division Dissertation Fellowship.
Jan. 2024: UCSB Abraham Wald Memorial Prize (ranking first in qualification exams).
Sept. 2022: UCSB Amazon Fellowship.
Sept. 2021: NYU GSAS Students for the Pathways to the Ph.D. Program.
May 2019: Dean's Undergraduate Research Fund (DURF), New York University Shanghai.
May 2019: PKU Academic Excellence Award.
May 2018: PKU Academic Excellence Award.
May 2018: PKU Scholarship for Outstanding Undergraduates.

INVITED TALKS & LECTURES

Nov. 2026: INFORMS Conference on Financial Engineering and Fintech, *Quantitative Finance and FinTech*, Stevens Institute of Technology, New Jersey, USA.
Feb. 2026: Mathematical Finance Colloquium, University of Southern California, California, USA.
Jan. 2026: Joint Mathematics Meetings, *Financial Mathematics: Theory and Practice*, Washington, D.C., USA.
Dec. 2025: Math & Machine Learning Seminar, California Institute of Technology, Pasadena, California, USA.
Oct. 2025: DECODE AI Annual Project Meeting, University of Texas at Austin, Austin, Texas, USA.
July 2025: *Optimal Transport and Applications*, University of California Santa Barbara, California, USA.
July 2025: SIAM Conference on Financial Mathematics and Engineering (FM 25), *Theory and Learning in Complex Stochastic System*, Miami, Florida, USA.
Mar. 2025: Western Conference on Mathematical Finance (WCMF 12), University of Southern California, Los Angeles, California, USA.
Feb. 2025: Center for Financial Mathematics and Actuarial Research (CFMAR) Graduate Student Research Lightning Talk, University of California Santa Barbara, California, USA.
Jan. 2025: Guest lecture on PyTorch, MATH 260L, University of California Santa Barbara, California, USA.
Nov. 2024: Banff International Research Station (BIRS) Workshop, *Modeling, Learning and Understanding: Modern Challenges between Financial Mathematics, Financial Technology and Financial Economics*, Banff, Alberta, Canada.
Oct. 2024: INFORMS Annual Meeting, *Learning and Applications of Mean-Field Models and Multi-Agent Systems*, Seattle, Washington, USA.
July 2024: *Optimal Transport through Midwest*, University of Wisconsin-Madison, Wisconsin, USA.
Nov. 2023: Center for Financial Mathematics and Actuarial Research (CFMAR) Graduate Student Research Lightning Talk, University of California Santa Barbara, California, USA.

ACADEMIC SERVICE

Referee for Peer-Reviewed Journals: *SIAM Journal on Control and Optimization (SICON)*, *Quantitative Finance*, *Digital Finance*, *IEEE Control Systems Letters (L-CSS)*.

Referee for Conferences: *IEEE Conference on Decision and Control (CDC) 2025–2026*, *American Control Conference (ACC) 2026*.

Organizer, UCSB Student Seminar in Financial Mathematics ([seminar website](#)), University of California Santa Barbara, California, USA. Dec. 2025 - June 2026

Organizer, Center for Financial Mathematics and Actuarial Research (CFMAR) Graduate Student Research Lightning Talk, University of California Santa Barbara, California, USA. Feb. 2025

Organizer, Reinforcement Learning Student Seminar, University of California Santa Barbara, California, USA. Jan. 2024 - June 2024

Session Chair, Southern California Quantitative Finance Forum, University of California Santa Barbara, California, USA. Apr. 2026

Panelist, Center for Financial Mathematics and Actuarial Research (CFMAR) Undergraduate Lab Career & Research Panel, University of California Santa Barbara, California, USA. Jan. 2026

TEACHING & MENTORING

Instructor, Princeton University: ORF 418 (Optimal Learning). Sept. 2026 - Dec. 2026

Undergraduate Research Mentor, Center for Financial Mathematics and Actuarial Research (CFMAR) Undergraduate Lab, University of California Santa Barbara, California, USA.

Project: *Financial Data Generation with Score-Based Diffusion Models* Sept. 2024 - Mar. 2025

Teaching Assistant, University of California Santa Barbara: PSTAT 213ABC (Graduate Stochastic Processes), PSTAT 120AB (Probability and Statistical Theory), PSTAT 170 (Introduction to Mathematical Finance). Sept. 2022 - May 2026

Teaching Assistant, New York University: MATH-UA 009 (Precalculus). Jan. 2022 - May 2022

WORKSHOPS & CONFERENCES ATTENDED

Nov. 2025: Southern California Quantitative Finance Forum, University of California Los Angeles, California, USA.

Feb. 2024: Institute for Mathematical and Statistical Innovation (IMSI) Workshop: *Decision Making and Uncertainty*, Chicago, Illinois, USA.

May 2022: Pacific Institute for the Mathematical Sciences-Centre de Recherches Mathématiques (PIMS-CRM) Probability Summer School, University of British Columbia (UBC), Vancouver, British Columbia, Canada.

ADDITIONAL SKILLS & ACTIVITIES

Languages: Chinese (native), English (fluent), Japanese (conversational), French (conversational).